MSBA 265 – SPECIAL ANALYTICS TOPICS

Road Accident Risk Prediction and Safety Prioritization

Final Project – Part 1



PREPARED BY
Anh Ngo

INSTITUTION
University of the Pacific

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1. Topic Examples

Several analytics approaches can support transportation safety decisions. The examples below illustrate how different methods could be applied to road-safety problems and the type of organizational decision each approach could support.

Topic Area	Real-World Example	Decision or Organizational Use
Supervised Machine Learning	Predict road accident risk using roadway, environmental, time-related, and historical conditions.	Help transportation agencies prioritize conditions for additional safety review.
Unsupervised Machine Learning	Group roadway observations with similar safety characteristics.	Identify clusters of road conditions that may benefit from similar interventions.
Dimensionality Reduction	Reduce many roadway and environmental variables into a smaller number of summary factors.	Simplify complex road-safety data for visualization and management reporting.
Natural Language Processing	Analyze crash narratives or incident descriptions.	Identify recurring crash causes, themes, or contributing factors.
Sentiment / Semantic Analysis	Analyze public comments about unsafe roads or transportation concerns.	Identify recurring public concerns that may warrant further investigation.
Social / Digital Channel Analytics	Analyze engagement with transportation-safety campaigns.	Help agencies understand which safety messages or outreach efforts receive the strongest response.

Selected focus: Supervised Machine Learning. This project uses labeled accident-risk outcomes and combines regression and classification to support a concrete transportation-safety prioritization decision.

2. Supervised Machine Learning in Transportation Safety

FROM ROAD CONDITIONS TO SAFETY PRIORITIZATION

Supervised Machine Learning in Transportation Safety

WHAT IT DOES

Supervised machine learning learns relationships from labeled historical data and applies those relationships to new observations. In road safety, it can estimate a continuous risk score or classify whether a condition belongs to a higher-risk group.

Labeled historical data → Learn relationships → New observations

REGRESSION
continuous risk score

CLASSIFICATION
higher-risk group

WHERE IT IS USED

Transportation safety analytics can use supervised models to estimate crash risk, classify crash severity, identify high-risk roadway conditions, and support safety prioritization.

- Crash-risk estimation
- Crash-severity classification
- High-risk roadway screening
- Safety-priority classification

WHY IT MATTERS

Road-safety agencies must allocate limited staff and funding across many possible needs. Data-driven safety analysis helps agencies identify higher-risk roadway features and prioritize safety reviews and potential investments more systematically *(FHWA, n.d.)*.

CONNECTION TO THIS PROJECT

1. **Multiple Linear Regression** estimates accident risk.
2. **Logistic Regression** identifies high-risk observations.
3. The combined output supports a transportation planner in deciding which road-condition profiles should receive higher priority for safety review.

PROJECT WORKFLOW

STEP 1 Road & Environmental Conditions → **STEP 2** Preprocessing → **STEP 3** MLR + Logistic Regression → **STEP 4** Risk Estimates & High-Risk Flags → **STEP 5** Safety Review Prioritization

> FHWA describes data-driven safety analysis as an evidence-based approach for identifying high-risk roadway features, supporting project decisions, and directing limited resources toward locations with greater safety-improvement potential.
>
> FEDERAL HIGHWAY ADMINISTRATION

SOURCES
FHWA. *Data-Driven Safety Analysis (DDSA).*
Kaggle. *Predicting Road Accident Risk.*
Chen et al. (2026). *Machine learning for traffic crash injury severity modelling.*

3. Project Proposal

3.1 Business Problem and Stakeholder

Transportation agencies must make safety decisions across many roadway conditions while operating with limited staff, time, and funding. A practical screening model can help identify road-condition profiles that appear more risky and therefore deserve closer review before resources are committed.

The primary intended users are transportation safety planners, traffic engineers, and road-safety analysts working for city or state transportation agencies. A road-safety consulting organization could also use the same type of tool when screening candidate conditions for a public-sector client.

> **Decision supported:** Which road-condition observations should receive higher priority for additional safety or engineering review?

The model is designed as decision support. It does not replace an engineer's professional judgment, field investigation, or formal roadway-safety analysis.

3.2 Project Question and Intended Use

> **Project question:** Can road, environmental, time-related, and historical characteristics be used to estimate accident risk and identify high-risk road conditions for transportation safety prioritization?

The intended final product is a Road Accident Risk Screening Tool. For a new set of roadway and environmental conditions, the tool would present a continuous estimated accident-risk score, a probability of belonging to a high-risk group, and a simple priority recommendation for review.

3.3 Success Measures

The project will be considered useful if the proposed models outperform simple baselines, produce stable and interpretable results on held-out validation data, and generate outputs that can be translated into a practical prioritization recommendation.

Component	Baseline	Evaluation Measures
Multiple Linear Regression	Training-set mean accident risk	RMSE, MAE, R^2
Logistic Regression	Majority-class prediction	Accuracy, precision, recall, F1, confusion matrix

4. Dataset and Planned Method

4.1 Dataset and Access

The project uses Kaggle's Predicting Road Accident Risk dataset from the 2025 Playground Series. The competition objective is to predict accident_risk, a continuous value between 0 and 1. Kaggle states that the competition data were generated from a deep-learning model trained on a simulated

road-accident dataset, so the data are synthetic rather than observations collected directly from a transportation agency.

Item	Project Data
Training set	517,754 rows × 14 columns
Kaggle test set	172,585 rows × 13 columns
Target	accident_risk (continuous, 0–1)
Missing values	None identified
Duplicate rows	None identified
License	CC BY 4.0, as listed by Kaggle

One row represents: one synthetic road-condition observation containing roadway, environmental, temporal, and historical accident characteristics.

Key fields: road_type, num_lanes, curvature, speed_limit, lighting, weather, road_signs_present, public_road, time_of_day, holiday, school_season, and num_reported_accidents. The id field is an identifier and is excluded from the predictor set.

Access and privacy: The data are available through Kaggle under the competition's access terms. The dataset does not contain personally identifying information.

> **Important limitation:** Because the dataset is synthetic and does not provide geographic coordinates or exact roadway locations, model outputs should be treated as a proof-of-concept screening approach. The project cannot identify a specific real-world intersection or claim causal relationships.

4.2 Planned Hybrid Modeling Approach

The project uses two complementary supervised-learning models:

Multiple Linear Regression

predicts the continuous accident_risk score. It is appropriate because the primary Kaggle target is continuous and the coefficients provide a transparent way to interpret how encoded predictors are associated with estimated risk.

Logistic Regression

provides a second decision-oriented output by classifying whether an observation belongs to a high-risk group. The proposed high-risk cutoff is the 75th percentile of accident_risk calculated from the training set only, avoiding information leakage from validation data.

For the classification component, recall will receive special attention because a false negative represents a truly high-risk observation that the screening process failed to flag.

4.3 Data Split and Evaluation Plan

The labeled Kaggle training dataset was split into approximately 80% training data and 20% validation data. The training set contains 414,203 observations, while the validation set contains 103,551 observations. The validation set is held out from model fitting and will be used to evaluate model performance and compare modeling decisions.

The separate Kaggle test.csv contains 172,585 observations but does not include accident_risk, so it will not be used for model evaluation. It will remain untouched until the final model is selected and will only be used to generate final predictions for Kaggle submission.

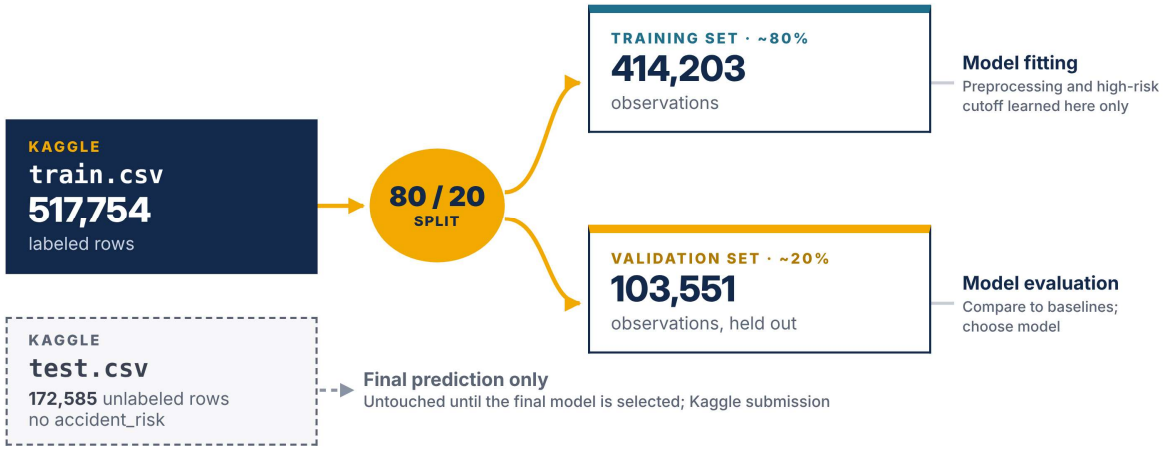


Figure 1. Data split and evaluation plan.

> **Leakage control:** All preprocessing parameters will be learned from the training data only and then applied to validation data to reduce data leakage.

5. Initial Data Exploration

Initial exploration focused on data quality, the distribution of the target, relationships between numeric predictors and accident risk, and differences in average risk across categorical conditions. The dataset is clean enough to proceed to model development without missing-value imputation.

5.1 Target Distribution

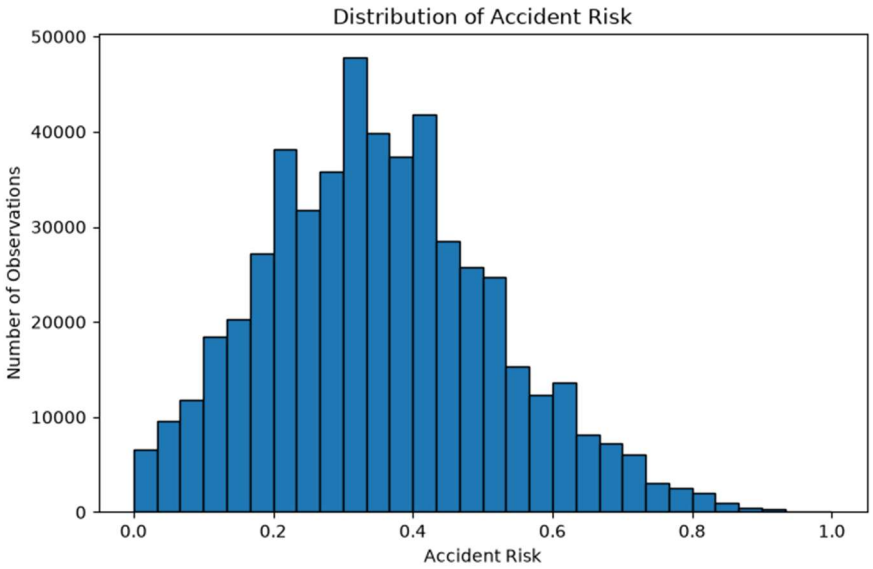


Figure 2. Distribution of accident_risk in the training data.

The target is continuous and concentrated around the middle of the 0–1 range rather than at the extremes. The training mean accident_risk is approximately 0.353, and the validation mean is approximately 0.351, indicating very similar target distributions after the 80/20 split.

5.2 Numeric Relationships

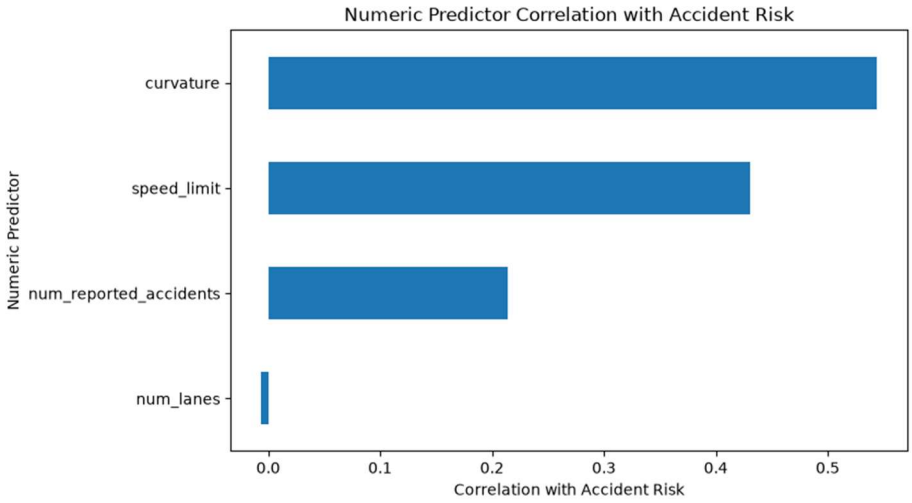


Figure 3. Numeric predictor relationships with accident_risk.

Curvature shows the strongest positive linear relationship with accident risk (r ≈ 0.544), followed by speed_limit (r ≈ 0.431). num_reported_accidents has a weaker positive relationship (r ≈ 0.214), while num_lanes shows almost no individual linear relationship (r ≈ -0.006). These patterns suggest that road geometry and speed-related characteristics may be especially useful for prediction, although correlation alone does not establish causation.

5.3 Lighting and Weather

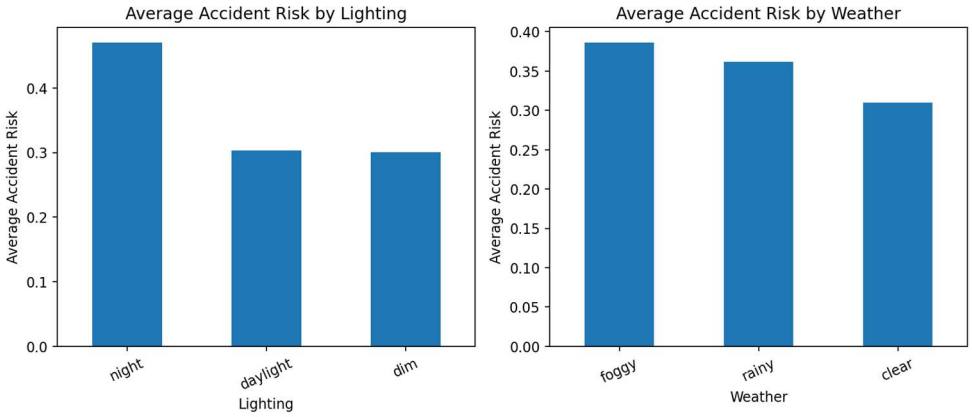


Figure 4. Average accident risk by lighting and weather condition.

Lighting and weather show meaningful differences in average risk. Night conditions have the highest average accident risk among lighting categories. Foggy conditions have the highest average risk among weather categories, while clear conditions have the lowest. These results support retaining both variables for model development.

5.4 Preparation Decisions

- Exclude id from the predictor set because it is an observation identifier.

- Encode categorical fields such as road_type, lighting, weather, and time_of_day before regression modeling.
- Convert Boolean predictors to model-ready numeric form.
- No missing-value imputation is currently required.
- Do not remove variables solely because they have weak individual correlations; evaluate their contribution in the multivariable models.
- Keep the Kaggle test set separate from model evaluation because its target values are not provided.

6. Feasibility and Next Steps

The project is feasible with the available data and course methods. The dataset is large, the target is well defined, the predictors include multiple types of road and environmental information, and the initial training/validation split preserves a similar target distribution. Several variables also show clear exploratory relationships with accident risk.

The main questions that remain are how much predictive improvement the models provide over simple baselines, whether the high-risk classification threshold produces a useful precision–recall balance, and whether the resulting outputs are reliable enough to support a practical screening recommendation.

Next steps:

1. Build the preprocessing pipeline for numeric, categorical, and Boolean predictors.
2. Establish the mean regression baseline and majority-class classification baseline.
3. Train and evaluate Multiple Linear Regression on the held-out validation set.
4. Create the high-risk target using a training-only cutoff and train Logistic Regression.
5. Compare regression and classification performance using the planned metrics.
6. Interpret model coefficients and error patterns in stakeholder terms.
7. Save the fitted preprocessing/model artifacts and verify that they reload in a fresh run.
8. Develop a simple Road Accident Risk Screening Tool if model performance is adequate.
9. Document limitations and define how a transportation planner should interpret the screening result.

7. Proposed Road Accident Risk Screening Tool

The final model outputs will be translated into a simple decision-support interface rather than presented only as technical metrics. A transportation safety planner could enter the available road and environmental conditions and receive a concise screening result.

Estimated Accident Risk	**0.62**
High-Risk Probability	**78%**
Priority Level	**HIGH**
Suggested Action	**Flag for additional transportation-safety review**

The example above illustrates the intended user experience; the numerical values are illustrative and will not be used as reported model results until the final models have been trained and evaluated.

8. References

Federal Highway Administration. (n.d.). *Data-Driven Safety Analysis (DDSA)*. U.S. Department of Transportation. https://highways.dot.gov/safety/data-analysis-tools/rsdp/data-driven-safety-analysis-ddsa

Kaggle. (2025). *Predicting Road Accident Risk: Playground Series – Season 5, Episode 10*. https://www.kaggle.com/competitions/playground-series-s5e10

Kaggle. (2025). *Predicting Road Accident Risk – Data*. https://www.kaggle.com/competitions/playground-series-s5e10/data

Chen, J., Meng, X., Chen, Z., Yuan, C., Zhao, Y., Zhang, J., et al. (2026). Machine learning for traffic crash injury severity modelling: A systematic review integrating text mining analysis. *European Transport Research Review, 18*, Article 96. https://doi.org/10.1186/s12544-026-00856-8

AI-Use Disclosure

Artificial intelligence tools were used as a support resource throughout this project. AI assisted with brainstorming the project structure, clarifying modeling concepts, troubleshooting Python code, improving documentation, and reviewing the clarity of written explanations.

AI was also used to support the visual presentation of the project, including the design and organization of the supervised machine learning infographic and the development of workflow diagrams. The underlying project concepts, model roles, workflow logic, and information presented in these visuals were reviewed and confirmed by the student.

All data exploration, model development, evaluation, and reported results are based on the project dataset and code executed by the student. AI-generated suggestions and visual content were reviewed, edited, and validated before being included in the project. AI was not used to create or modify the dataset, fabricate model results, or replace the student's interpretation of the analysis.

Final decisions regarding the business problem, modeling approach, evaluation plan, visual presentation, and interpretation of results remain the responsibility of the student.